

Ngan Hien Lyle, MD, MDS

Infectious Diseases Physician | Data Scientist

Kelowna, British Columbia, Canada | ngan.lyle@ubc.ca | nganlyle.github.io

GitHub | ORCID: 0000-0002-7162-9127

Profile

Infectious diseases physician and data scientist with interests at the intersection of clinical medicine, wearable physiological sensing, continuous waveform data, and artificial intelligence. My research focuses on developing and applying data-driven methods to understand physiological changes during illness and to make high-quality physiological data more accessible for research.

Current Appointments

Clinical Assistant Professor	2022–Present	University of British Columbia
Infectious Diseases Consultant	2014–Present	Kelowna General Hospital, Kelowna, BC
Flex Advisor, Southern Medical Program	2024–Present	University of British Columbia

Research Interests

- Wearable physiological sensing and continuous physiological monitoring
- Photoplethysmography (PPG) and physiological waveform analysis
- Artificial intelligence, machine learning, and modern statistical methods
- Physiological changes associated with infectious diseases
- Open data, reproducible and participatory research

Current Work

Sympath

Sympath is an emerging research platform focused on making high-quality physiological data easier to collect, analyze, and share. The project combines wearable physiological sensors, mobile data collection, waveform analysis, and open research infrastructure.

www.sympath.ca

Education

Master of Data Science	2019–2020	University of British Columbia, Kelowna, BC
M.D.C.M., Medicine	2002–2006	McGill University, Montréal, QC
B.Sc. Honours, Chemistry	1997–2001	Queen's University, Kingston, ON

Medical Training & Credentials

Fellowship, Infectious Diseases	2010–2012	University of British Columbia
Residency, Internal Medicine	2007–2010	University of Colorado Health Sciences Center
Diploma in Tropical Medicine and Hygiene	2012	Gorgas Institute, Lima, Peru

Royal College of Physicians and Surgeons of Canada

- Fellow of the Royal College of Physicians and Surgeons of Canada (FRCPC), Infectious Diseases, 2013
- Fellow of the Royal College of Physicians and Surgeons of Canada (FRCPC), Internal Medicine, 2012

Selected Research Experience

PPG Monitoring in the Management of Severe Dengue

2022–2023

Oxford University Clinical Research Unit, Ho Chi Minh City, Vietnam

- Research fellow working with Prof. Yacoub on the application of artificial intelligence to continuous PPG data in patients with dengue.
- Developed data-driven approaches to characterize physiological changes associated with dengue shock syndrome.

Modeling and Visualization of COVID-19 Outbreaks in Canada

2020–2022

University of British Columbia

- Applied open epidemiological, census, and long-term care data to investigate patterns of COVID-19 transmission in British Columbia and Ontario.
- Developed computational approaches for analysis and visualization of population-level infectious disease data.

Sepsis Immunopathogenesis

2013–2014

University of British Columbia

- Research investigating host immune responses and biomarkers associated with sepsis and organ dysfunction.

Interferon- α Responses in Lymphoid Tissue in HIV

2009–2010

University of Colorado

- Investigated sex-related differences in innate immune responses in HIV infection.

Clinical Experience

Infectious Diseases Consultant

2014–Present

Kelowna General Hospital, Kelowna, BC

- Inpatient and outpatient infectious diseases consultation
- Outpatient parenteral antimicrobial therapy (OPAT)

Locum Infectious Diseases Consultant

2019–2022

Hospitals across British Columbia, including Lions Gate Hospital, Royal Inland Hospital, and University Hospital of Northern British Columbia

Physician, Tuberculosis Services 2022

BC Centre for Disease Control

Infectious Diseases Consultant 2014

Vancouver General Hospital, Vancouver, BC

Teaching

Flex Advisor, Southern Medical Program 2024–Present

University of British Columbia

- Advise medical students on scholarly and research projects.

Clinical Assistant Professor, Department of Medicine 2022–Present

University of British Columbia

- Supervision and teaching of medical students and residents.
- Large-group teaching in MED 422.

Clinical Instructor, Department of Medicine 2015–2022

University of British Columbia

Publications

1. Lyle NN, et al. An artificial intelligence-based approach to identify volume status in patients with severe dengue using wearable PPG data. *PLOS Digital Health*. 2025;4(7).
2. Ho CQ, et al., Lyle NN. Kinetics of cardiovascular and inflammatory biomarkers in paediatric dengue shock syndrome. *Oxford Open Immunology*. 2024;5(1):iqae005.
3. Ho CQ, et al., Lyle NN. Applying artificial intelligence and digital health technologies, Viet Nam. *Bulletin of the World Health Organization*. 2023;101:487–492.
4. Pena OM, et al., Lyle NH. An endotoxin tolerance signature predicts sepsis and organ dysfunction at initial clinical presentation. *EBioMedicine*. 2014;1(1):64–71.
5. Meditz A, et al., Lyle NH. CCR5 expression is reduced in lymph nodes of HIV-1-infected women compared to men, but does not mediate sex differences in viral loads. *Journal of Infectious Diseases*. 2014;209(6):922–930.
6. Lyle NH, et al. Barriers to the effective treatment of sepsis: antimicrobial agents, sepsis definitions, and host-directed therapies. *Annals of the New York Academy of Sciences*. 2014;1323:101–114.
7. Xia J, et al., Lyle NH. INVEX—a web-based tool for integrative visualization of expression data. *Bioinformatics*. 2013;29(24):3232–3234.

8. Lyle N, Boyd J. The potential for PCR based testing to improve diagnosis and treatment of sepsis. *Current Infectious Disease Reports*. 2013;15(5):372–379.

Selected Presentations

Potential Utility of Continuous PPG Monitoring in Management of Severe Dengue

ASTMH Annual Meeting, October 2023

Modeling and Visualization of COVID-19 Outbreak in Ontario

All Things Open, Statistics Canada, June 2020

Awards & Funding

- CANSSI Rapid Response Program, 2020 — \$6,000
- Master of Data Science Domestic Scholarship, University of British Columbia, 2019
- Faculty of Science Graduate Award, University of British Columbia, 2012–2014
- Clinician Investigator Program Fellowship, University of British Columbia, 2013–2015 — \$120,000
- AMMI Canada / Pfizer Post-Residency Fellowship, 2012 — \$60,000
- Dr. Stefan Gryzbowski Prize, University of British Columbia, 2011
- CROI Young Investigator Award, 2010

Leadership & Service

Physician Lead, Nutrition Project

2018

Kelowna General Hospital

- Led a Facility Engagement project focused on improving access to nutritious food for physicians and healthcare workers after hours.

Ambulatory Care Executive

2017–2019

Kelowna General Hospital

Antimicrobial Stewardship Committee

2016–2019

Kelowna General Hospital

Technical Skills

Programming Python, R

Data science Statistical analysis, machine learning, physiological waveform analysis, data visualization

Other SQL, L^AT_EX

Updated September 2026